

Chapter 8: Advanced Theories of Covalent Bonding

T. K. Deskins, Ph.D.

Professor of Physics & Chemistry
Southern Union State Community College

PHY201
Lecture 8



Part I

Why We Need Advanced Bonding Theories

Outline of Part I: Why We Need Advanced Bonding Theories

Limitations of Lewis Structures and VSEPR

Outline of Part I: Why We Need Advanced Bonding Theories

Limitations of Lewis Structures and VSEPR

What Lewis Structures Cannot Explain

Review: What We Know So Far

- ▶ Lewis structures show **how many** bonds exist and where lone pairs sit.
- ▶ VSEPR theory predicts **molecular geometry** from electron-pair repulsions.

What These Models Fail to Explain

- ▶ **Why** bond angles are 109.5° , 120° , or 180° at the orbital level
- ▶ Why O_2 is attracted to a magnet (paramagnetic), even though its Lewis structure shows no unpaired electrons
- ▶ The **electronic structure** of molecules with delocalized electrons

This chapter introduces three complementary tools: **valence bond (VB) theory**, **orbital hybridization**, and **molecular orbital (MO) theory**.

The O₂ Paradox: A Magnetic Mystery

The Lewis Structure of O₂



Lewis structure predicts **all electrons paired** $\Rightarrow$ O₂ should be **diamagnetic**.

Only **MO theory** can fully explain this observation.

We will return to it in Section 8.4.

The Experimental Reality

Liquid O₂ is **visibly attracted** to a magnet — it is **paramagnetic**.

Paramagnetism requires **unpaired electrons**.

Part II

Valence Bond Theory

Outline of Part II: Valence Bond Theory

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Valence Bond Theory

What Is Valence Bond Theory?

Core Idea of Valence Bond (VB) Theory

A **covalent bond** forms when:

1. An orbital on one atom **overlaps** an orbital on a second atom, *and*
2. Each orbital contains **one electron** — the two electrons pair up in the shared region.

Example: H₂ Formation

- ▶ Each H atom has one electron in a 1s orbital.
- ▶ As two H atoms approach, the 1s orbitals **overlap**.
- ▶ The shared electron pair is concentrated **between the nuclei** — this is the bond.

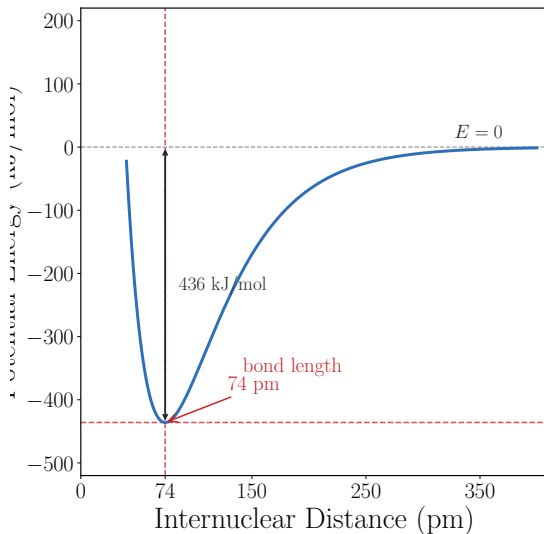
Greater orbital overlap $\Rightarrow$ stronger bond. Overlap is maximized along specific

Bond Formation: Energy vs. Internuclear Distance

Potential Energy Curve for H_2

- ▶ At large separation: atoms are independent; energy = 0.
- ▶ As atoms approach: attractive forces lower energy.
- ▶ At **equilibrium bond length** (74 pm): energy is minimum (most stable).
- ▶ Closer than 74 pm: nuclear repulsion raises energy sharply.

The **bond dissociation energy** (436 kJ/mol for H_2) equals the



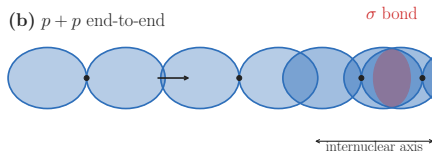
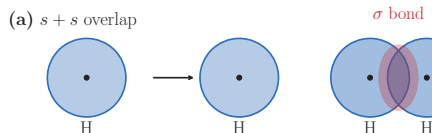
Sigma Bonds (σ)

Definition: Sigma Bond (σ bond)

A **sigma bond** is a covalent bond in which electron density is concentrated **along the internuclear axis** (end-to-end overlap).

How σ Bonds Form

- ▶ $s + s$ overlap (e.g., H-H)
- ▶ $s + p$ overlap (e.g., H-F)
- ▶ $p + p$ end-to-end overlap
- ▶ Hybrid + hybrid overlap (most C-C, C-N, etc.)



Pi Bonds (π)

Definition: Pi Bond (π bond)

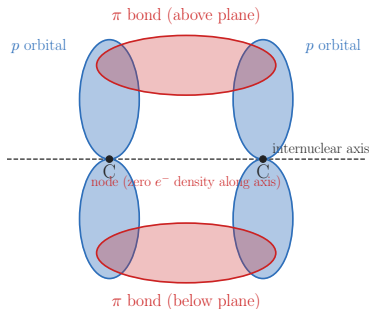
A **pi bond** is formed by the **side-by-side** (lateral) overlap of two *p* orbitals. Electron density lies *above and below* the internuclear axis, with a **node** along the axis itself.

Multiple Bond Composition

- ▶ Single bond: **1 σ**
- ▶ Double bond: **1 σ + 1 π**
- ▶ Triple bond: **1 σ + 2 π**

Key Distinction

Pi Bond (π): Side-by-Side *p* Overlap



Check Your Understanding

For each molecule below, state the number of σ bonds and π bonds:

- (a) N_2 (triple bond)
- (b) CO_2 (two double bonds)
- (c) H_2O (two single bonds)

Come to lecture to see the answer.

Part III

Hybrid Atomic Orbitals

Outline of Part III: Hybrid Atomic Orbitals

Hybrid Atomic Orbitals

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Hybrid Atomic Orbitals

Why Hybridization?

The Problem with Pure Atomic Orbitals

Carbon's electron configuration is $1s^2 2s^2 2p^2$.

Pure p orbitals are at 90° to each other — but CH_4 bond angles are **109.5°** . How?

Hybridization: The Solution

Hybridization is the mathematical combination (**linear combination**) of atomic orbitals to form new **hybrid orbitals** that:

- ▶ Have **shapes distinct** from original s and p orbitals
- ▶ Are **equivalent in energy and shape** to each other
- ▶ Number equal to the orbitals combined
- ▶ Point toward bonding partners — **maximizing overlap**

Hybridization only occurs in **bonded** atoms. Isolated atoms do not have hybrid

How to Determine Hybridization

Strategy: Count Electron Domains on the Central Atom

1. Draw the Lewis structure.
2. Count **electron domains** (groups) on the central atom:
bonds (single, double, or triple each count as **one domain**) + lone pairs.
3. Match the number to the hybridization:

Domains	Hybridization	Geometry	Example
2	sp	linear	BeCl_2 , CO_2
3	sp^2	trigonal planar	BF_3 , C_2H_4
4	sp^3	tetrahedral	CH_4 , H_2O
5	$sp^3 d$	trig. bipyramidal	PCl_5
6	$sp^3 d^2$	octahedral	SF_6

sp Hybridization: Linear Geometry

Formation of *sp* Hybrid Orbitals

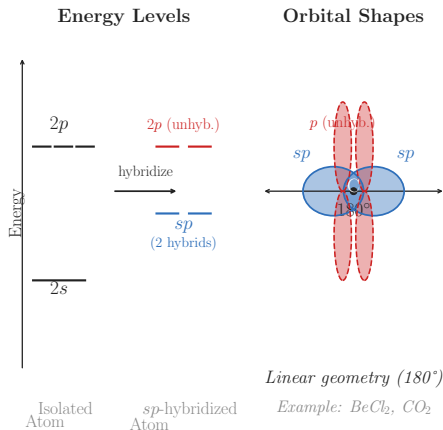
Mixing **one** *s* + **one** *p* orbital produces **two** *sp* hybrids:

- ▶ Two lobes point in **opposite directions** (180° apart)
- ▶ Each hybrid points along the internuclear axis
- ▶ Two unhybridized *p* orbitals remain, perpendicular to the axis

Example: BeCl_2

Be: $[\text{He}] 2s^2 \xrightarrow{\text{hybridize}} 2 \text{ } sp \text{ orbitals}$

Each *sp* orbital overlaps a Cl $3p$ orbital



sp^2 Hybridization: Trigonal Planar Geometry

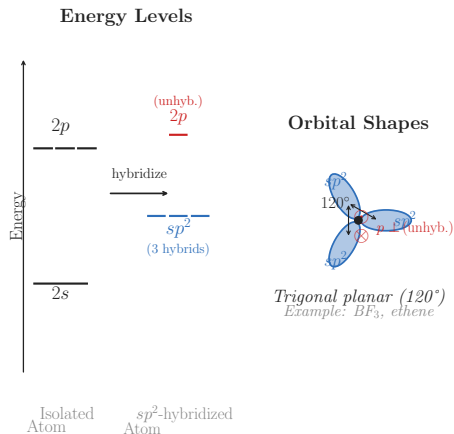
Formation of sp^2 Hybrid Orbitals

Mixing **one** s + **two** p orbitals produces **three** sp^2 hybrids:

- ▶ Three lobes in a **trigonal planar** arrangement (120°)
- ▶ One unhybridized p orbital remains, perpendicular to the plane
- ▶ This p orbital is available for π bonding

Examples: BF_3 , C_2H_4

Each B in BF_3 : 3 sp^2 orbitals $\Rightarrow$ 3 σ bonds at 120° .



sp^3 Hybridization: Tetrahedral Geometry

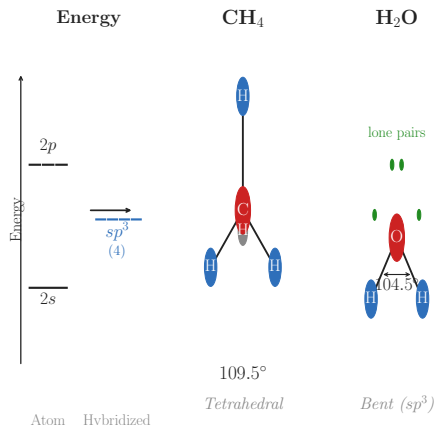
Formation of sp^3 Hybrid Orbitals

Mixing **one** s + **three** p orbitals produces **four** sp^3 hybrids:

- ▶ Four lobes point toward the corners of a **tetrahedron**
- ▶ Bond angles: **109.5°**
- ▶ No unhybridized p orbitals remain

Examples

- ▶ CH_4 : 4 sp^3 hybrid orbitals $\Rightarrow$ 4 σ bonds at 109.5°
- ▶ NH_3 : 3 sp^3 bonds + 1 sp^3 lone pair $\Rightarrow$



Check Your Understanding

Determine the hybridization of the **central atom** in each molecule:

- (a) SO_3 (trigonal planar, no lone pairs on S)
- (b) PF_3 (pyramidal shape)
- (c) XeF_2 (linear molecule, 3 lone pairs on Xe)

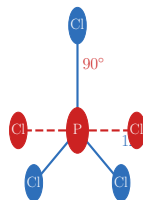
Come to lecture to see the answer.

sp^3d and sp^3d^2 Hybridization

sp^3d : Trigonal Bipyramidal (5 domains)

- ▶ Mix $s + 3p + d \Rightarrow$ **5 hybrids**
- ▶ 3 equatorial (120°) + 2 axial (90° from equatorial)
- ▶ Example: PCl_5 , SF_4 , ClF_3 , XeF_2

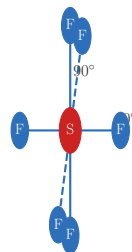
sp^3d : Trig. Bipyramidal
 Example: PCl_5



● Equatorial (120°)

● Axial (90°)

sp^3d^2 : Octahedral
 Example: SF_6



All bond angles: 90°

sp^3d^2 : Octahedral (6 domains)

- ▶ Mix $s + 3p + 2d \Rightarrow$ **6 hybrids**
- ▶ All bonds at 90° — perfectly symmetric
- ▶ Example: SF_6 , PCl_6^- , XeF_4

Part IV

Multiple Bonds

Outline of Part IV: Multiple Bonds

Multiple Bonds

Outline of Part IV: Multiple Bonds

Multiple Bonds

Ethene (C₂H₄): A Double Bond

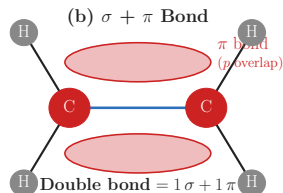
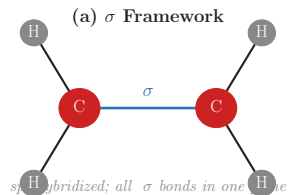
sp^2 Hybridization in Ethene

Each carbon in C₂H₄ is sp^2 hybridized:

- ▶ Three sp^2 orbitals form σ bonds: C–C and two C–H
- ▶ One unhybridized p **orbital** on each C overlaps **side-by-side** $\Rightarrow \pi$ bond
- ▶ Molecule is **planar** (all atoms in one plane)

C₂H₄ double bond = 1 σ + 1 π

The π bond forces planarity: rotation would break the side-by-side p overlap.



Ethyne (C_2H_2): A Triple Bond

sp Hybridization in Ethyne

Each carbon in C_2H_2 is sp hybridized:

- ▶ Two sp orbitals: C–H σ bond + C–C σ bond (180° apart)
- ▶ **Two** unhybridized p orbitals on each C
- ▶ These four p orbitals overlap side-by-side $\Rightarrow$ **two π bonds**

Result: Linear Molecule



(a) σ Framework (linear, sp)

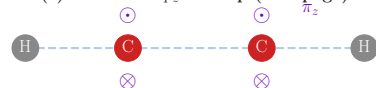


180° — linear; all sp σ bonds along the axis

(b) π Bond 1: p_y overlap π_y



(c) π Bond 2: p_z overlap (into page) π_z



Triple bond = $1 \sigma + 2 \pi$ (cylindrical π cloud)

Restricted Rotation About Double Bonds

Why π Bonds Restrict Rotation

Rotation about a double bond would require **breaking the π bond**: the p orbitals must remain parallel to overlap. Bond energies show the energy cost:

Bond	Length (pm)	Energy (kJ/mol)
C–C (single)	150.6	347
C=C (double)	133.5	614
C $\equiv$ C (triple)	120.8	839

Geometric Isomers: A Consequence

The restricted rotation about C=C means **cis** and **trans** isomers of alkenes are distinct molecules with different properties — they cannot interconvert at room temperature.

Benzene and Electron Delocalization

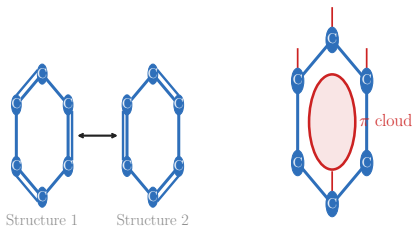
Benzene: sp^2 Carbon Ring

Each of the 6 carbon atoms in C_6H_6 is sp^2 hybridized:

- ▶ 3 sp^2 orbitals form σ bonds (C–C and C–H)
- ▶ 1 unhybridized p orbital on each C, all perpendicular to the ring
- ▶ All 6 p orbitals overlap to form a **delocalized π system**

Electron density is **spread equally** over the entire ring — neither resonance

Resonance Structures True π Bonding: Delocalized



Neither structure alone is correct.

π electrons are delocalized.

All C–C bonds equal (139 pm)

sp^2 C atoms; one p per C

6 p orbitals form delocalized π system

Hybridization and Resonance: A Key Rule

Assigning Hybridization in Resonance Structures

When determining hybridization, count σ bonds, lone pairs, and unpaired electrons only — the arrangement of π bonds does not matter.

Example: Ozone (O_3)

- ▶ O_3 has 2 resonance structures, each showing one $\text{O}=\text{O}$ and one $\text{O}-\text{O}$.
- ▶ Central O: 2 σ bonds + 1 lone pair = 3 domains $\Rightarrow sp^2$
- ▶ The π electrons are **delocalized** over both $\text{O}-\text{O}$ bonds.

Rule: hybridization of the central atom is the **same in all resonance forms** of a molecule.

Check Your Understanding

For **formaldehyde** ($\text{H}_2\text{C}=\text{O}$):

- (a) What is the hybridization of the carbon atom?
- (b) How many σ bonds and π bonds does the molecule have?
- (c) What is the $\text{H}-\text{C}-\text{H}$ bond angle?

Come to lecture to see the answer.

Part V

Molecular Orbital Theory

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Limitations of Valence Bond Theory

What VB Theory Cannot Explain

- ▶ VB theory localizes electrons **between pairs of atoms** — but many molecules have electrons **delocalized over the whole molecule**.
- ▶ VB theory cannot predict the **paramagnetic behavior of O_2** .
- ▶ VB theory assigns electrons to specific bonds — MO theory lets them occupy the whole molecule.

Molecular Orbital (MO) Theory

Molecular orbital theory treats electrons as **delocalized** throughout the entire molecule, occupying orbitals that span all atomic nuclei. These molecular orbitals arise from the **linear combination of atomic orbitals (LCAO)**.

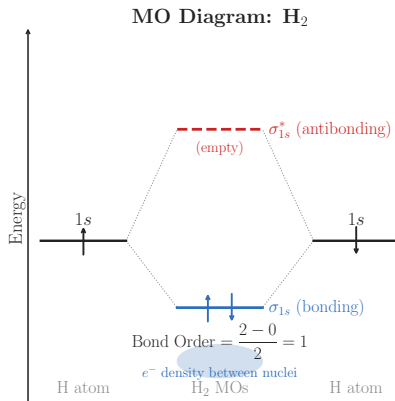
Bonding and Antibonding Orbitals

How MOs Form from AOs

When two atomic orbitals combine, they produce **two molecular orbitals**:

- ▶ **Bonding MO** (σ , π): constructive interference $\Rightarrow$ electron density **between** nuclei; **lower** energy than original AOs.
- ▶ **Antibonding MO** (σ^* , π^*): destructive interference $\Rightarrow$ a **node** **between** nuclei; **higher** energy.

Electrons in **bonding MOs** stabilize the molecule.



Bond Order

Definition: Bond Order

Bond order is a measure of how many net bonding pairs exist between two atoms:

$$\text{Bond Order} = \frac{(\text{bonding electrons}) - (\text{antibonding electrons})}{2}$$

Interpreting Bond Order

- ▶ Bond order = 0: **no stable bond** (molecule does not exist)
- ▶ Bond order = 1: single bond
- ▶ Bond order = 2: double bond
- ▶ Bond order = 3: triple bond
- ▶ **Higher bond order** $\Rightarrow$ shorter bond length and larger bond energy

H₂ Example

He₂: Why Helium Does Not Dimerize

MO Configuration of He₂

He atom: $1s^2$ — so He₂ would have 4 electrons total.

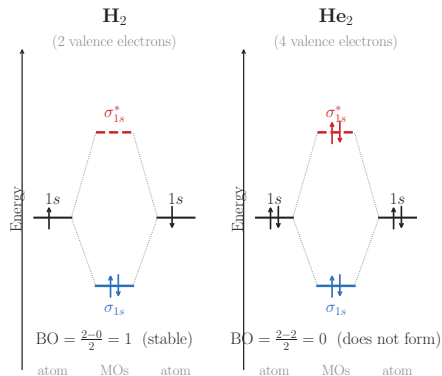
MO filling: $(\sigma_{1s})^2 (\sigma_{1s}^*)^2$

$$\text{Bond Order} = \frac{2 - 2}{2} = \boxed{0}$$

Conclusion

Bond order = 0 $\Rightarrow$ He₂ is **not a stable molecule**.

The bonding and antibonding orbitals **cancel each other out** exactly.



Check Your Understanding

The **hydrogen molecule ion** H_2^+ has only **one electron**.

- (a) Write its MO electron configuration.
- (b) Calculate the bond order.
- (c) Is H_2^+ stable? Is it more or less stable than H_2 ?

Come to lecture to see the answer.

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

MO Theory for Second-Period Diatomics

Orbitals Involved

For Period-2 homonuclear diatomics (Li_2 through Ne_2), both $2s$ and $2p$ atomic orbitals contribute MOs:

- ▶ $2s$ AOs $\Rightarrow \sigma_{2s}$ and σ_{2s}^*
- ▶ $2p$ AOs (end-to-end) $\Rightarrow \sigma_{2p}$ and σ_{2p}^*
- ▶ $2p$ AOs (side-by-side) $\Rightarrow$ two π_{2p} and two π_{2p}^*

s-p Mixing

For **lighter** Period-2 elements (Li_2 through N_2), the σ_{2s} and σ_{2p} orbital energies are **close enough to mix**, causing the π_{2p} orbitals to be **lower in energy** than σ_{2p} .

For O_2 , F_2 , Ne_2 : no s - p mixing; σ_{2p} is **lower** than π_{2p} .

MO Energy Diagrams: O₂ and N₂

MO Filling Rules

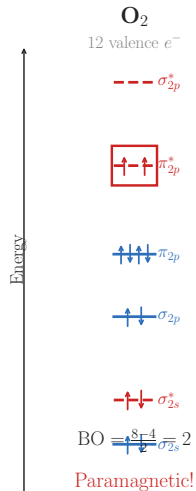
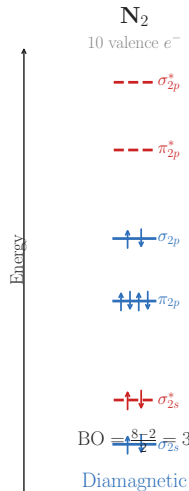
Fill MOs by **Aufbau** (lowest energy first), **Hund's rule** (degenerate orbitals fill singly first), and **Pauli exclusion**.

N₂ (10 valence electrons)

$$(\sigma_{2s})^2(\sigma_{2s}^*)^2(\pi_{2p})^4(\sigma_{2p})^2$$

$$\text{Bond Order} = \frac{8 - 2}{2} = \boxed{3} \quad (\text{triple bond})$$

All electrons paired $\Rightarrow$
diamagnetic



O₂ Paramagnetism: The Mystery Solved

Why O₂ Is Paramagnetic

The two highest-energy electrons in O₂ occupy the **degenerate** π_{2p}^* **orbitals** one each (Hund's rule) $\Rightarrow$ **two unpaired electrons**.

Unpaired electrons have a magnetic moment $\Rightarrow$ the molecule is attracted to magnetic fields.

Lewis Structure vs. MO Theory

- ▶ Lewis structure: O=O (all paired) — **predicts diamagnetic**
- ▶ MO theory: 2 unpaired e^- in π^* — **predicts paramagnetic**

This is a fundamental victory of MO theory over simpler bonding models. It correctly predicts a measurable, real-world property that Lewis theory gets **completely wrong**.

Summary: Properties of Second-Period Diatomics

Molecule	Valence e^-	Bond Order	Unpaired e^-	Magnetism
Li ₂	2	1	0	diamagnetic
Be ₂	4	0	0	(does not exist)
B ₂	6	1	2	paramagnetic
C ₂	8	2	0	diamagnetic
N ₂	10	3	0	diamagnetic
O ₂	12	2	2	paramagnetic
F ₂	14	1	0	diamagnetic
Ne ₂	16	0	0	(does not exist)

Pattern: Bond order peaks at N₂ (the strongest bond in chemistry); it decreases on either side.

Diamagnetic vs. Paramagnetic: Definitions

Paramagnetism

A substance is **paramagnetic** if it contains **unpaired electrons**. Paramagnetic materials are *attracted* to an external magnetic field.

Diamagnetism

A substance is **diamagnetic** if **all electrons are paired**. Diamagnetic materials are weakly *repelled* by a magnetic field.



Test: Count unpaired electrons in the MO diagram.

Any unpaired electron $\Rightarrow$ paramagnetic.

All paired $\Rightarrow$ diamagnetic.

Check Your Understanding

Using MO theory, determine for F_2 :

- (a) The bond order.
- (b) Whether F_2 is paramagnetic or diamagnetic.
- (c) How many σ and π bonds does F_2 have?

Come to lecture to see the answer.

Outline of Part V: Molecular Orbital Theory

Molecular Orbital Theory

MO Theory: Second-Period Diatomics

Comparing Bonding Theories

Three Theories: A Comparison

Feature	VB Theory	MO Theory
Electron location	localized between atom pairs	delocalized over whole molecule
Bond types	σ and π	bonding and anti-bonding MOs
Geometry	via hybridization	via electron count in MOs
Paramagnetism	cannot predict	predicts correctly
Resonance	multiple structures needed	single wave function
Best for	predicting geometry, hybridization	predicting magnetic and electronic properties

Key Terms and Summary

VB Theory & Hybridization

- ▶ **Valence bond theory:** bonds via orbital overlap
- ▶ **σ bond:** end-to-end overlap; all single bonds
- ▶ **π bond:** side-by-side overlap; 2nd/3rd bonds
- ▶ **Hybridization:** mixing AOs to form hybrid orbitals
- ▶ sp , sp^2 , sp^3 , sp^3d , sp^3d^2 : hybrid types

MO Theory

- ▶ **MO theory:** electrons delocalized over molecule
- ▶ **Bonding MO:** stabilizes molecule (lower energy)
- ▶ **Antibonding MO (*):** destabilizes (higher energy)
- ▶ **Bond order:** $\frac{\text{bonding} - \text{antibonding}}{2}$
- ▶ **Paramagnetic:** unpaired electrons
- ▶ **Diamagnetic:** all electrons paired

Final Check Your Understanding

Without drawing a full MO diagram, predict whether each species is **stable**, **paramagnetic**, and give its **bond order**:

(a) O_2^{2-} (peroxide ion — add 2 electrons to O_2)

(b) N_2^+ (remove 1 electron from N_2)

Come to lecture to see the answer.